

Preserving and improving the legacy of eScience: the GLOBO experience

<https://git.isac.cnr.it/montella/globone-glowpp.git>

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Abstract

Legacy scientific codes are essential to long-standing operational workflows but often cannot fully exploit modern heterogeneous high-performance computing (HPC) architectures. GLOBO, a global atmospheric circulation model developed at CNR-ISAC, is one such model that is still used for research and operational weather forecasting. In this work, we present an ongoing refactoring of GLOBO to improve its scalability and prepare it for GPU acceleration and cloud execution. The modernization focuses on three main areas: dynamic memory allocation, replacement of legacy point-to-point MPI communications with optimized collective operations, and OpenACC-based GPU offloading. We describe the redesign of the communication strategy, which replaces multiple MPI_Isend/MPI_Irecv exchanges with collective primitives (MPI_Bcast, MPI_Scatterv, MPI_Sendrecv, MPI_Reduce) to reduce overhead and improve parallel efficiency. Scalability experiments across multiple spatial resolutions show that a new collective communication strategy consistently matches or exceeds the original implementation, delivering up to 40% runtime reduction in high-resolution, multi-node configurations. These results lay the groundwork for refactoring GLOBO toward a fully dynamic, GPU-accelerated, cloud-ready execution, preserving its scientific heritage while ensuring sustainable performance on future HPC platforms.

Introduction

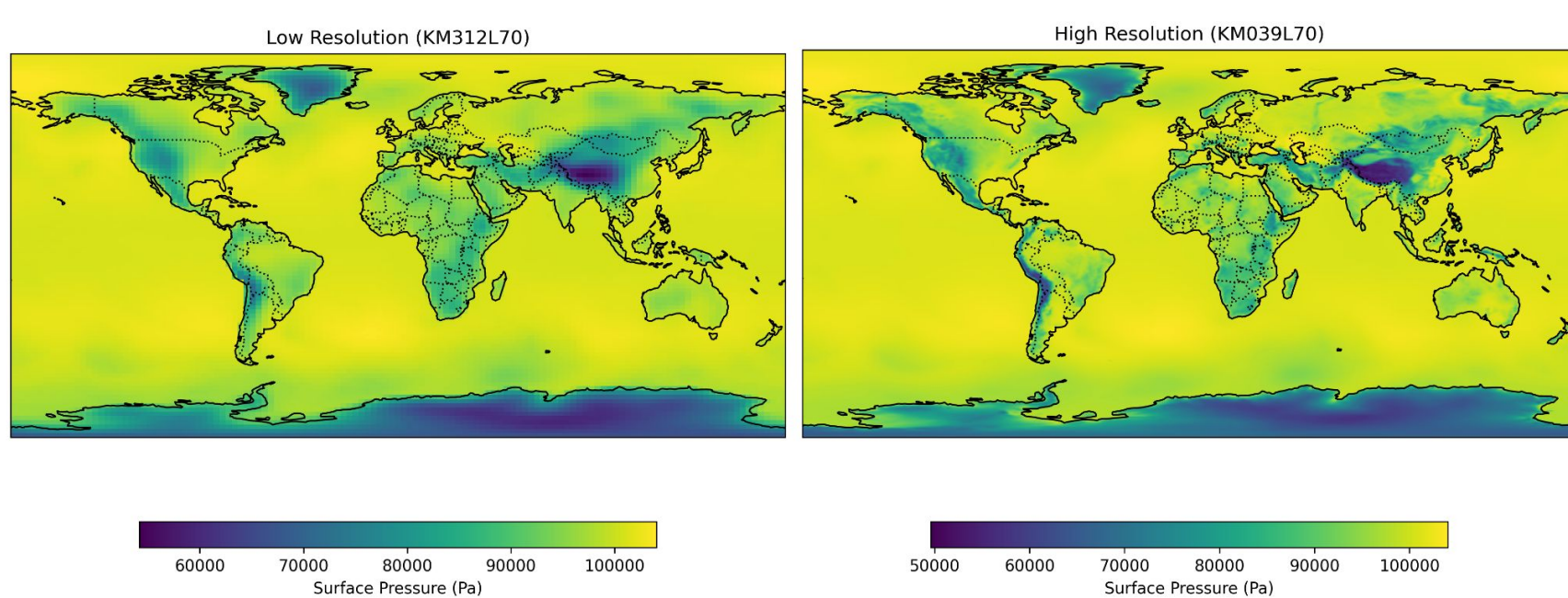
Modernizing legacy scientific codes is a central challenge in the era of heterogeneous high-performance computing (HPC). Many weather and climate models, originally developed for CPU-only architectures, lack the modularity and parallel scalability required to leverage today's multi-GPU systems.

The GLOWPP project, aims to refactor and optimize a global weather prediction model for sustainable use on both HPC clusters and cloud platforms. The goal is to bridge the gap between traditional science and modern computing by combining reproducibility, portability, and performance.

The GLOBO Model

GLOBO is a global, hydrostatic atmospheric general circulation model developed at CNR-ISAC. It originates from the Bologna Limited Area Model (BOLAM) and is still used today both for research on predictability and for operational forecasting. As part of the CNR-ISAC modeling chain, GLOBO produces **daily 7-day global forecasts** and runs a **40-member ensemble** every week to generate sub-seasonal probabilistic forecasts, contributing to the **S2S initiative**. The model is based on a **hydrostatic dynamical core** on a regular latitude-longitude Arakawa C-grid, with explicit time-stepping. Thanks to its flexible design, GLOBO can run at multiple horizontal resolutions (from **312 km down to 19 km**) and vertical resolutions ranging from **50 to 70 levels**.

Comparison of Surface Pressure at Different Resolutions



Design and Implementation Strategy

The modernization of GLOBO follows three main development lines:

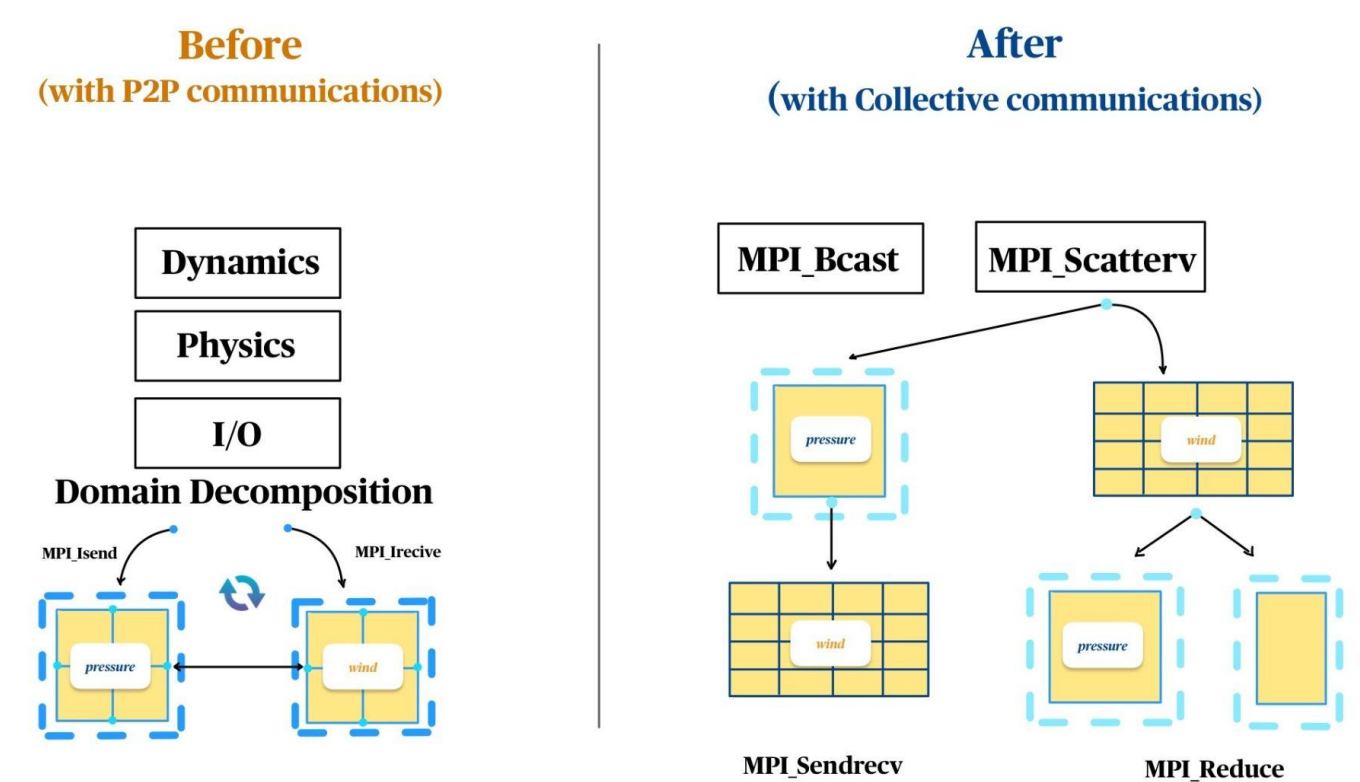
- **Dynamic memory allocation** → reduces static resource usage and increases flexibility in domain decomposition.
- **Optimized MPI collective operations** → replace legacy point-to-point communication to improve scalability on multi-node systems.
- **GPU preparation with OpenACC** → enables future offloading of computationally intensive kernels.

Current work focuses on **communication restructuring**. In the legacy implementation, the root process distributed data through multiple MPI_Isend/MPI_Irecv calls followed by MPI_Wait, causing **bottlenecks and synchronization overhead** at high resolution. The new strategy leverages **collective operations**:

- **MPI_Bcast** → global parameter dissemination
- **MPI_Scatterv** → efficient sub-domain distribution
- **MPI_Sendrecv** → simultaneous boundary exchange
- **MPI_Reduce** → collective reduction of global fields

This redesign eliminates sequential loops, simplifies code maintenance, and **improves scalability** across multi-node HPC environments. It also lays the foundation for **dynamic allocation and GPU acceleration**.

Globo Model Communication

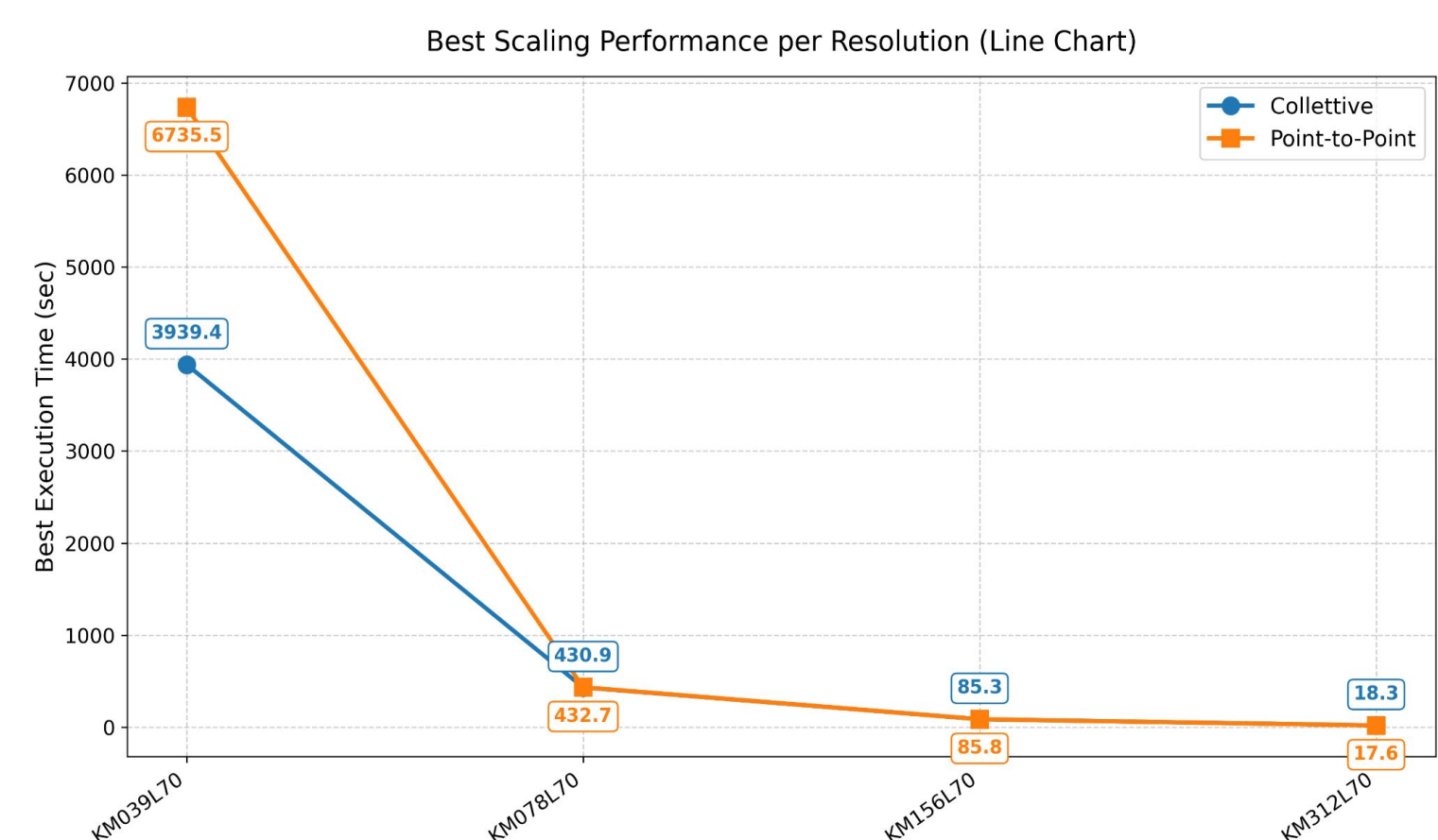


Early Results and Evaluation

We tested the impact of replacing **point-to-point communication** with **MPI collectives** across four spatial resolutions.

- **Coarse resolutions (KM312L70, KM156L70):** both methods perform similarly, as computation dominates.
- **Intermediate resolution (KM078L70):** collectives maintain stable execution times up to 192 processes, while point-to-point shows variability and slower scaling.
- **High resolution (KM039L70):** collectives reduce runtime from **6,736 s to 3,939 s** (~40% faster), clearly improving scalability.

Overall, **collective operations improve communication efficiency** in GLOBO at large scale, reducing overhead and synchronization costs.



Conclusion

- **Simplified code:** collective MPI reduces complexity compared to multiple point-to-point exchanges.
- **Improved scalability:** experiments show consistent or superior performance, with large gains at high resolution.
- **Next steps:** optimize ghost-boundary communications, introduce **dynamic memory management** and **GPU offloading**.

These efforts establish the foundation for a **production-ready GLOBO**, capable of efficient execution on **heterogeneous HPC and cloud platforms**.

